

GumBall6900

The index fund
that chooses itself

An immutable signal-directed onchain fund with continuously clearing GBX mining, tenure-locked miner rates, oracleless acquisitions, and in-kind redemption.

20M

GENESIS GBX

1-16

MINING SLOTS

80 / 20

HANDOFF / RESONANCE

Fixed

TENURE RATE

READ THIS FIRST

Development design, not a deployed system.

GumBall6900 is experimental software. It is not deployed, has not received an independent external audit, and is not authorized for user funds. Exact Mine economics, target-chain dependencies, legal provenance, and a signed deployment manifest remain unresolved.

This edition supersedes the daily pooled mining design. The Solidity and tested economic fixtures are authoritative. A local green build is engineering evidence, not a safety or release claim.

EDITION

v0.4, 12 August 2026. Not deployed and not authorized for user funds. Independent external audit not completed.

Genesis issuance	20,000,000 GBX for canonical liquidity
Later issuance	One permanently bound immutable Mine
Supply limit	No economic cap; ERC20Votes retains a uint208 safety ceiling
Governance	Resonance actions plus increase-only Mine capacity
Legal status	Farplace, give.fun, Liquid Signal, and transitive lineage remain unresolved release blockers

ORIENTATION

Contents

01	Economic loop	04	04	Supply and redemption	07
02	Mining market	05	05	Governance and risks	08
03	Fixed-tenure fairness	06			

PART I

01 The economic loop

Mining brings USDG in; sGBX signals direct each interval of a seven-day stream.

- 01 A participant takes a mining slot at its current hourly decaying USDG price.
- 02 The incumbent accrues GBX continuously at the fixed rate assigned on entry.
- 03 Twenty percent of a nonempty-slot handoff routes through Resonance; eighty percent becomes a displaced-miner claim.
- 04 ResonanceRouter holds sub-threshold USDG; a qualifying balance restarts a seven-day stream under live signals.
- 05 Strategies pull released USDG and exchange it for assets whose complete payments become Fund liabilities.
- 06 A GBX holder may burn GBX for a selected pro-rata basket of raw Fund assets.

EMPTY SLOTS

A first occupation routes 100% of its USDG payment because no displaced miner exists.

NO TEAM FEE

Mining has no team, founder, management, or protocol fee.

PART II

02 A continuously clearing market

Each independently replaceable slot falls from its opening USDG price to zero in one hour.

A miner buys the right to accrue GBX until replacement and the possibility of receiving 80% of the next payment. The second value is uncertain: without a successor, no handoff claim arrives. This rollover risk disciplines the price miners are willing to pay.

After each handoff, the next opening price is the paid amount multiplied by an immutable factor, with an immutable minimum. Expected GBX value, market liquidity, gas, replacement probability, and contract risk all influence the real clearing price.

MOMENT	CONTRACT OUTCOME
Start of hour	Price equals the slot opening price
Thirty minutes	Approximately half remains
One hour or later	Price is zero; replacement still allowed
Nonempty replacement	80% claim / 20% Resonance

NOT GUARANTEED

Profitability, a successor payment, GBX liquidity, and frequent replacement are market outcomes—not contract guarantees.

CALLER BOUNDS

Every purchase supplies an expected epoch, a deadline, and a maximum price, so a miner front-run at purchase cannot be charged the reset opening price.

PART III

03 Miners keep the rate they bought

A slot rate changes only when that slot changes hands.

Checkpoints, cumulative-mining thresholds, redemptions, and capacity increases never rewrite an occupied slot's GBX-per-second rate.

STATE	SLOT 0 INCUMBENT	NEW SLOTS
Capacity 1; global 100 GBX/hour	100 GBX/hour	Closed
Capacity grows to 3	Still 100 GBX/hour	Open
New occupations	Still 100 GBX/hour	About 33 GBX/hour each

This prevents governance from changing the economic deal after entry. Only future occupations divide the current global rate by current capacity. A threshold crossing similarly changes only the rate offered at a later handoff.

ACCEPTED TRADEOFF

Aggregate issuance can temporarily exceed the current global rate while old high-rate tenures remain.

PART IV

04 Accrued mining counts before redemption

Fund crystallizes every live slot before taking the common supply snapshot.

GBX begins with 20,000,000 genesis tokens. Its only later issuer is the permanently bound Mine. Global rates offered to future occupants halve at immutable cumulative-mining thresholds, then continue at a positive tail on every modeled horizon. ERC20Votes' uint208 ceiling remains the remote implementation bound.

Rewards accrue continuously but mint at checkpoints. Fund calls checkpointAll before every redemption, so already-earned GBX cannot be omitted from the denominator.

PAYOUT

$$\text{floor}(\text{Fund token balance} \times \text{GBX burned} \div \text{post-checkpoint pre-burn supply})$$

BOUNDED WORK

Capacity is permanently capped at 16; checkpoint cost grows linearly but cannot become unbounded.

OMISSIONS

Assets a redeemer leaves out remain permanently for the post-redemption supply.

PART V

05 Immutable by design

The timelock can maintain membership and add mining concurrency; it cannot rewrite mining tenures.

TIMELOCKED ACTIONS

- Add a Strategy
- Permanently kill a Strategy
- Register a Bribe reward token, up to eight
- Increase Mine capacity, from one to at most sixteen

ABSENT POWERS

- No capacity decrease or incumbent repricing
- No emission setter or replacement authority
- No proxy, migration, rescue, pause, or arbitrary executor
- No Fund withdrawal or liquidity NFT recovery

Major risks include uncertain mining demand and GBX liquidity, rollover risk, temporarily elevated issuance after expansion, rapid signal movement, failing token dependencies, permanent deployment mistakes, and immutable custody.

RELEASE STATUS

Independent audit, parameter review, provenance clearance, and signed deployment evidence are still required.